



A Firm-Level Microsimulation for VAT Policy Analysis

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Motivation



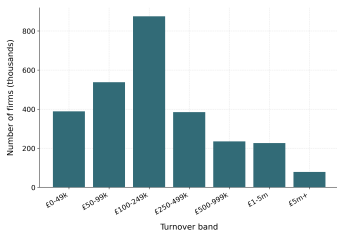
How should the UK **cost reforms** to its VAT registration threshold — changes to its **level**, **shape**, or **rate** — on a **single, common, reproducible basis**,

when the firm-level data needed are not openly available?

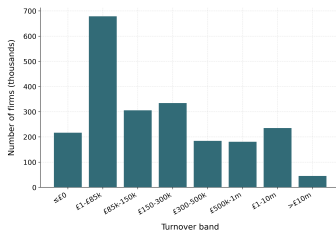
- The UK VAT threshold is a **notch, not a kink**: cross it and 20% falls due on the firm's *entire* turnover — not just the slice above.
- A discrete liability jump of $\tau T^* = \text{£}17,000$ ($= 20\% \times \text{£}85,000$) at the threshold.
- Threshold was **£85,000** (2023–24), raised to **£90,000** on 1 April 2024 — after a seven-year nominal freeze, the longest in the tax's history.
- **Among the highest thresholds in the OECD**, and it sits in a *dense* part of the firm-size distribution.
- Firms respond by **bunching just below** it; the cliff-edge keeps the threshold under recurrent pressure for reform.

Firms cluster just below the threshold

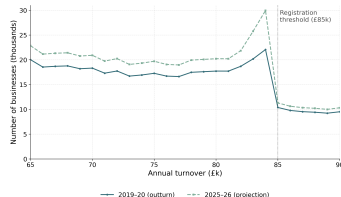
Cross £85,000 and 20% VAT falls due on the firm's *entire* turnover — a £17,000 jump — so firms cluster right below it.



All UK firms by turnover band
(ONS)



VAT-registered firms by band
(HMRC)



Bunching below £85k (OBR, 2023)

- An **open, reproducible firm-level microsimulation** that costs UK VAT threshold reforms on a common static basis.
- Built on **synthetic firm microdata** calibrated to published HMRC + ONS aggregates — **released openly** and regenerable under any counterfactual schedule.
- Prices three reform families — **level** (move the threshold), **shape** (a graduated taper), **rate** (a reduced-rate band) — **three ways**:
 1. **statically** (mechanical reclassification);
 2. by the exact, elasticity-free **dominated-region** distortion each removes;
 3. **behaviourally**, conditional on an assumed turnover elasticity e .
- **Headline**: only the graduated taper removes the notch's distortion; raising the threshold is the dearest, least efficient option.

- **Bunching / notch methodology** — source of the “dominated region”
 - Saez (2010); Kleven and Waseem (2013); Kleven (2016)
- **VAT registration threshold (empirical)**
 - Liu et al. (2021) (UK notch, $\sim 43\%$ voluntary registration); Liu, Lockwood and Tam (2024); Onji (2009); Asatryan and Peichl (2017); Waseem (2024)
- **Does bunching identify a structural elasticity?**
 - Blomquist et al. (2021); Bertanha et al. (2023) — *not* non-parametrically identified
- **Structural models of size-based thresholds**
 - Garicano et al. (2016); Gourio and Roys (2014)

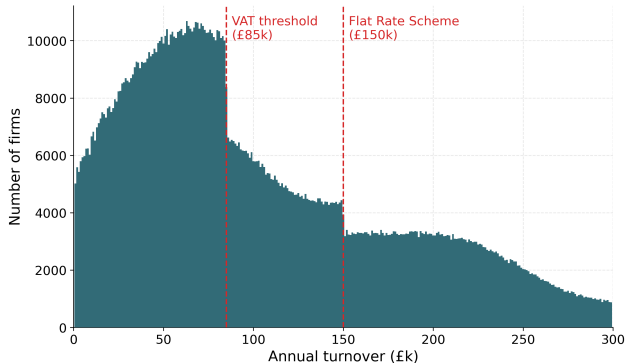
Data and calibration



- **No open firm-level VAT microdata exist** $\Rightarrow$ build a synthetic population calibrated to published aggregates (data year 2023–24).
- **Calibration targets:**
 - ONS *UK Business: Activity, Size and Location* — firm counts by sector $\times$ turnover band
 - HMRC *VAT Annual Statistics* — registered counts and net VAT liability by band / sector
- **Scale:** $\approx 2.94\text{m}$ firm records, weighted to represent $\approx 2.5\text{m}$ UK firms.
- **Accuracy:** $\approx 90\%$ across five calibrated dimensions.
- Approach mirrors open household tax-benefit microsimulation (Ghenis et al., 2022).

► how firms are built

► distribution



Density steps down at the £85,000 registration threshold and again at the £150,000 Flat Rate Scheme ceiling. The step is inherited from coarse HMRC band targets — quantified later by a placebo test.

Static costing



Static revenue at threshold T

$$R(T) = \sum_i w_i v_i \mathbf{1}\{y_i \geq T\}.$$

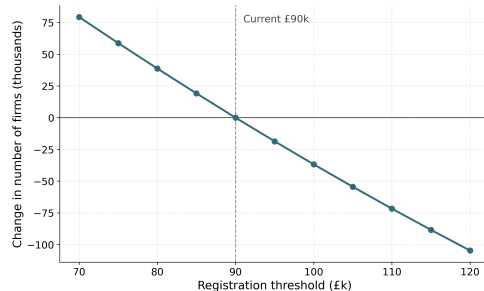
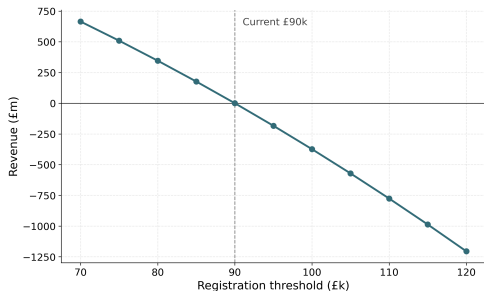
- Turnover held fixed $\Rightarrow$ revenue moves **only through firms whose registration status flips** (those between the old and new cutoff).
- Three reform **levers**, four schedules, all on one base:
 - **Level** — move the threshold up or down
 - **Shape** — graduated taper, 0% at £85k $\rightarrow$ 20% at £105k
 - **Rate** — reduced-rate band (10% or 15%) in £85k–£105k
- For each, recompute $R = \sum_i w_i v_i$ under the new schedule and difference.

Validation: the April-2024 anchor reform



- Cost the £85k→£90k April-2024 rise, my estimate vs HMRC's published costing.
- 2025–26: £175m vs £185m — agree within £7–21m per year across the horizon.
- Both turn to a small *gain* by 2028–29 as fiscal drag lifts the frozen baseline.
- Microdata are calibrated to HMRC aggregates ⇒ a **consistency check**, not out-of-sample validation.

Static sweep: cost of moving the threshold (2025–26)



From a £90k base (£203.8bn), the schedule is roughly linear, $\approx$ £150–220m per £5k step: e.g. –£374m to raise to £100k, +£665m to lower to £70k. Even a £30k rise to £120k costs $\sim$ £1.2bn — under 0.6% of the base.

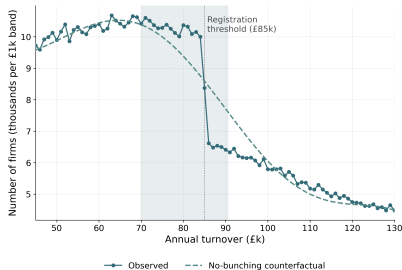
Reform	Lever	Static	Dominated region
Raise to £100k	Level	—£508m	relocated
Taper £85–105k	Shape	—£336m	removed
Reduced 10%	Rate	—£343m	split in two
Reduced 15%	Rate	—£171m	split in two

Common £85,000-notch / £183.6bn 2023–24 base. The level move is the most expensive yet removes no distortion; the taper removes it entirely for less.

“Split in two” = a second notch appears where the reduced-rate band ends (£105k reverts to 20%), so the empty band is fragmented, not shrunk.

The notch's distortion



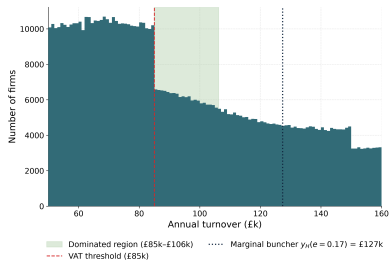


Excess mass over a polynomial no-bunching counterfactual g :

$$b = \frac{B(y^*)}{g(y^*)} \approx \frac{\Delta y^*}{y^*}, \quad E = \sum_{\text{window}} (n_j - c_j).$$

- Reproduced density step: ratio $b = 0.060$ (SE 0.005), excess mass $E = 8,712$ firms (SE 662).
- Placebo: remove the step from the calibration target $\Rightarrow E$ collapses to 0–99 firms.
- So the step is inherited from coarse HMRC bands, not firm behaviour — the generator has no location choice.
- Genuine threshold bunching remains the administrative fact of Liu et al. (2021).

The dominated region: an exact, elasticity-free distortion



- At T^* , profit jumps *down* by $\tau T^* = £17,000$ — VAT on turnover already earned.
- The **dominated region** is the band just above T^* where no firm locates (strictly worse than at T^*):

$$a = T^* \frac{\tau}{1 - \tau} = \text{£21,250}.$$

The notch in the firm's profit:

$$\pi(y) = \begin{cases} y - c(y), & y < T^*, \\ (1 - \tau)y - c(y), & y \geq T^*. \end{cases}$$

- Interval (£85,000, £106,250), $\approx 137,000$ weighted firms.
- Depends on τ and T^* **alone** — no elasticity, no cost specification. An order of magnitude wider than a kink's $\sim £1k$.

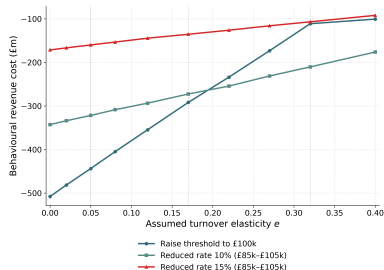
- $$a' = T_1 \frac{\tau - r}{1 - \tau}; \quad a + a' = \text{£}21,562 \text{ (15\%)}, \quad \text{£}22,569 \text{ (10\%)}. \quad \square$$

- **Graduated taper** — continuous, no notch and no dominated region: it removes the distortion entirely.

15/22

Behavioural layer





Each firm rescales turnover under the reform:

$$y_i^* = y_i \left(\frac{1 - \tau_{\text{reform}}}{1 - \tau_{\text{base}}} \right)^e, \quad \frac{\partial \ln y}{\partial \ln(1 - \tau)} = e.$$

- Iso-elastic simulator re-optimises each firm (left); the elasticity e is **assumed**, not identified.
- Sweep $e \in \{0.05, 0.17, 0.32\}$. At $e = 0.17$ base-broadening offsets much of each static loss:
 - raise to £100k: £508m → **£292m**
 - 10% band: £343m → **£273m**
 - 15% band: £171m → **£135m**
- Static cost is the $e \rightarrow 0$ limit; **only $e = 0.05$ has an external anchor** (Kleven and Waseem, 2013).
- Taper **excluded**: its marginal-rate channel is not credibly identified iso-elastically.

Conclusion



- **What this is:** an open, reproducible firm-level microsimulation that models the UK VAT threshold as a turnover-tax notch on synthetic firm data, and prices level / shape / rate reforms **three ways**.
- **Static:** reproduces HMRC's anchor costing within $\sim$ £10m; raising to £100k costs £508m, the taper £336m.
- **Distortion:** the dominated region $a = £21,250$ is exact and elasticity-free — **only the taper removes it**; the level move merely relocates it; a reduced rate splits it in two.
- **Transparency:** synthetic bunching is mechanical (placebo: $8,712 \rightarrow \sim 99$ firms) — no elasticity is read off the data.
- **Behavioural offsets** are conditional on assumed e ; the static cost and dominated region are the elasticity-free anchors.
- **Next:** identify e from administrative microdata; add welfare / MVPF accounting; sectoral incidence.

Thank you!

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Scan: paper, data & code

Appendix: how the synthetic firms are built

- Draw firms by sector $\times$ ONS turnover band; turnover smoothed within band:

$$y_i = \underline{b} + (\bar{b} - \underline{b}) u_i + \varepsilon_i, \quad u_i \sim \mathcal{U}(0, 1), \quad \varepsilon_i \sim \mathcal{N}(0, \sigma_b^2).$$

- Input share $x_i = \rho_i y_i$ ($\rho_i \in [0.6, 1.5]$, rescaled Beta) $\Rightarrow$ net VAT base $v_i = y_i - x_i$.
- Each firm gets a strictly positive weight $w_i = e^{\theta_i}$; weighted target $\hat{T}_k(\theta) = \sum_i a_{ki} w_i$.
- Fit θ by **gradient descent** on a symmetric relative-error loss:

$$L(\theta) = \frac{1}{K} \sum_k \lambda_k \min\{(\hat{T}_k/T_k - 1)^2, (T_k/\hat{T}_k - 1)^2\} + \frac{\alpha}{N} \sum_i |\theta_i|.$$

- The threshold is a **generator parameter**: every counterfactual — higher cutoff, taper, reduced rate — regenerates a fresh, internally consistent population.

Appendix: notch vs. kink

- A **kink** taxes only the turnover *above* the threshold — the marginal rate jumps, but liability is continuous.
- A **notch** taxes the firm's *entire* turnover once it registers — liability jumps discretely by $\tau T^* = \text{£}17,000$.
- The UK VAT threshold is a notch $\Rightarrow$ a strict **dominated region** exists (no firm should locate just above T^*), whereas a kink leaves only a $\sim\text{£}1\text{k}$ smoothed gap.
- This is why the distortion can be measured **exactly and without any elasticity**.

Appendix: behavioural costs across e

Reform	Static ($e=0$)	$e=0.05$	$e=0.17$	$e=0.32$
Raise to £100k	−£508m	−£444m	−£292m	−£111m
Reduced 10% band	−£343m	−£322m	−£273m	−£210m
Reduced 15% band	−£171m	−£160m	−£135m	−£107m

A larger e makes every flat reform cheaper (base-broadening). The graduated taper is omitted — its base-broadening and marginal-rate channels cannot both be netted in the iso-elastic cost. The $e \rightarrow 0$ limit reproduces the static costs to within $\sim$ £1m.

► back

Appendix: secondary-notch arithmetic (reduced rate)

A reduced rate r in $[T^*, T_1]$ with $T_1 = \text{£}105,000$:

- **Primary** notch shrinks: jump rT^* , width $a = T^* \frac{r}{1-r} = \text{£}15,000$ (15%), $\text{£}9,444$ (10%).
- **Secondary** notch at the band top, where rate reverts to τ : jump $(\tau - r)T_1 = \text{£}5,250$ (15%), width $a' = T_1 \frac{\tau - r}{1-\tau} = \text{£}6,562$ (15%), $\text{£}13,125$ (10%).
- **Total** dominated width $a + a' = \text{£}21,562$ (15%), $\text{£}22,569$ (10%) — both marginally *above* the $\text{£}21,250$ baseline.

The reduced rate splits the empty band; it does not shrink it.

Appendix: limitations and future work

- **Turnover-tax approximation** — real VAT is on value added with input reclaim; ~43% of below-threshold firms register voluntarily (Liu et al., 2021), so the effective notch is smaller.
- **Partial efficiency metric** — dominated-region width indexes misallocation only, omitting compliance-cost savings (Keen and Mintz, 2004); a full welfare / MVPF accounting is needed.
- **Aggregate only** — no sectoral incidence (VAT liability by sector excluded from the optimiser).
- **Synthetic, partial-equilibrium** — abstracts from price pass-through; no genuine firm location choice.